28. ADC - Analog to Digital Converter

28.1. Features

- 10-bit Resolution
- 0.5 LSB Integral Non-Linearity
- ±2 LSB Absolute Accuracy
- 13 - 260µs Conversion Time
- Up to 76.9kSPS (Up to 15kSPS at Maximum Resolution)
- Six Multiplexed Single Ended Input Channels
- Two Additional Multiplexed Single Ended Input Channels (TQFP and VFQFN Package only)
- Temperature Sensor Input Channel
- Optional Left Adjustment for ADC Result Readout
- 0 - V_{CC} ADC Input Voltage Range
- Selectable 1.1V ADC Reference Voltage
- Free Running or Single Conversion Mode
- Interrupt on ADC Conversion Complete
- Sleep Mode Noise Canceler

28.2. Overview

The device features a 10-bit successive approximation ADC. The ADC is connected to an 8-channel Analog Multiplexer which allows eight single-ended voltage inputs constructed from the pins of Port A. The single-ended voltage inputs refer to 0V (GND).

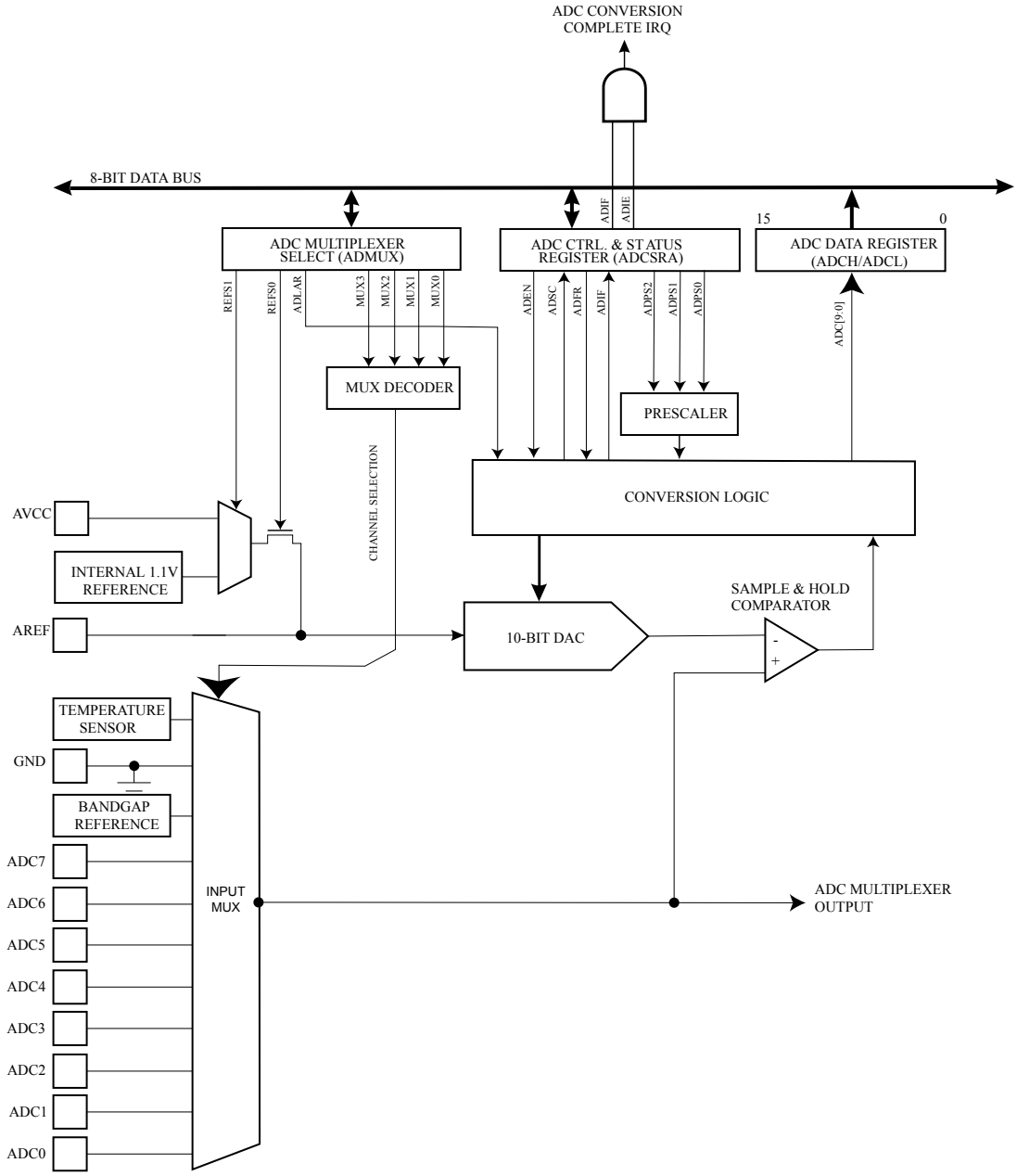
The ADC contains a Sample and Hold circuit which ensures that the input voltage to the ADC is held at a constant level during conversion. A block diagram of the ADC is shown below.

The ADC has a separate analog supply voltage pin, AV_{CC}. AV_{CC} must not differ more than ±0.3V from V_{CC}. See section ADC Noise Canceler on how to connect this pin.

The Power Reduction ADC bit in the Power Reduction Register (PRR.PRADC) must be written to '0' in order to be enable the ADC.

The ADC converts an analog input voltage to a 10-bit digital value through successive approximation. The minimum value represents GND and the maximum value represents the voltage on the AREF pin minus 1 LSB. Optionally, AV_{CC} or an internal 1.1V reference voltage may be connected to the AREF pin by writing to the REFSn bits in the ADMUX Register. The internal voltage reference must be decoupled by an external capacitor at the AREF pin to improve noise immunity.

Figure 28-1. Analog to Digital Converter Block Schematic Operation



The analog input channel is selected by writing to the MUX bits in the ADC Multiplexer Selection register ADMUX.MUX[3:0]. Any of the ADC input pins, as well as GND and a fixed bandgap voltage reference, can be selected as single ended inputs to the ADC. The ADC is enabled by writing a '1' to the ADC Enable bit in the ADC Control and Status Register A (ADCSRA.ADEN). Voltage reference and input channel selections will not take effect until ADEN is set. The ADC does not consume power when ADEN is cleared, so it is recommended to switch off the ADC before entering power saving sleep modes.

The ADC generates a 10-bit result which is presented in the ADC Data Registers, ADCH and ADCL. By default, the result is presented right adjusted, but can optionally be presented left adjusted by setting the ADC Left Adjust Result bit ADMUX.ADLAR.

If the result is left adjusted and no more than 8-bit precision is required, it is sufficient to read ADCH. Otherwise, ADCL must be read first, then ADCH, to ensure that the content of the Data Registers belongs to the same conversion: Once ADCL is read, ADC access to Data Registers is blocked. This means that if

ADCL has been read, and a second conversion completes before ADCH is read, neither register is updated and the result from the second conversion is lost. When ADCH is read, ADC access to the ADCH and ADCL Registers is re-enabled.

The ADC has its own interrupt which can be triggered when a conversion completes. When ADC access to the Data Registers is prohibited between reading of ADCH and ADCL, the interrupt will trigger even if the result is lost.

Related Links

Power Management and Sleep Modes on page 62
Power Reduction Register on page 65

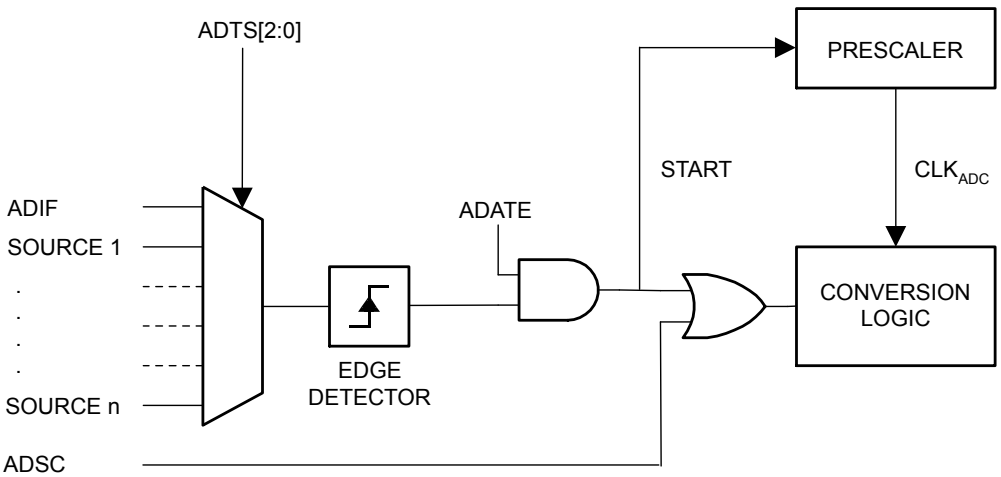
28.3. Starting a Conversion

A single conversion is started by writing a '0' to the Power Reduction ADC bit in the Power Reduction Register (PRR.PRADC), and writing a '1' to the ADC Start Conversion bit in the ADC Control and Status Register A (ADCSRA.ADSC). ADCS will stay high as long as the conversion is in progress, and will be cleared by hardware when the conversion is completed. If a different data channel is selected while a conversion is in progress, the ADC will finish the current conversion before performing the channel change.

Alternatively, a conversion can be triggered automatically by various sources. Auto Triggering is enabled by setting the ADC Auto Trigger Enable bit (ADCSRA.ADATE). The trigger source is selected by setting the ADC Trigger Select bits in the ADC Control and Status Register B (ADCSRB.ADTS). See the description of the ADCSRB.ADTS for a list of available trigger sources.

When a positive edge occurs on the selected trigger signal, the ADC prescaler is reset and a conversion is started. This provides a method of starting conversions at fixed intervals. If the trigger signal still is set when the conversion completes, a new conversion will not be started. If another positive edge occurs on the trigger signal during conversion, the edge will be ignored. Note that an interrupt flag will be set even if the specific interrupt is disabled or the Global Interrupt Enable bit in the AVR Status REgister (SREG.I) is cleared. A conversion can thus be triggered without causing an interrupt. However, the Interrupt Flag must be cleared in order to trigger a new conversion at the next interrupt event.

Figure 28-2. ADC Auto Trigger Logic



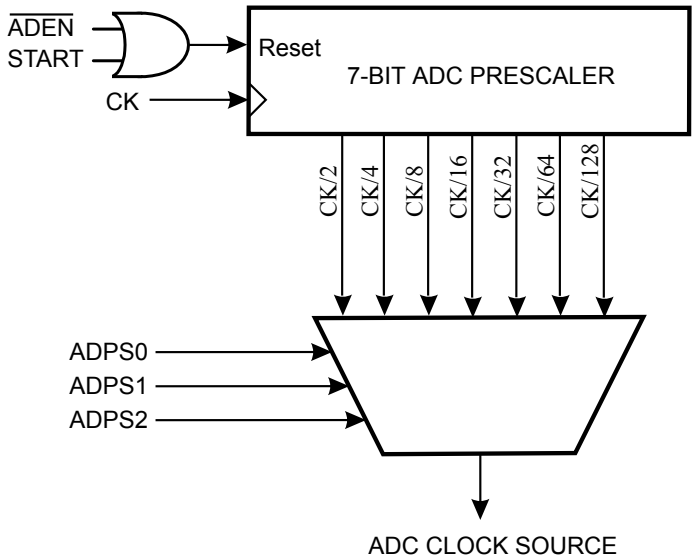
Using the ADC Interrupt Flag as a trigger source makes the ADC start a new conversion as soon as the ongoing conversion has finished. The ADC then operates in Free Running mode, constantly sampling and updating the ADC Data Register. The first conversion must be started by writing a '1' to

ADCSRA.ADSC. In this mode the ADC will perform successive conversions independently of whether the ADC Interrupt Flag (ADIF) is cleared or not.

If Auto Triggering is enabled, single conversions can be started by writing ADCSRA.ADSC to '1'. ADSC can also be used to determine if a conversion is in progress. The ADSC bit will be read as '1' during a conversion, independently of how the conversion was started.

28.4. Prescaling and Conversion Timing

Figure 28-3. ADC Prescaler



By default, the successive approximation circuitry requires an input clock frequency between 50kHz and 200kHz to get maximum resolution. If a lower resolution than 10 bits is needed, the input clock frequency to the ADC can be higher than 200kHz to get a higher sample rate.

The ADC module contains a prescaler, which generates an acceptable ADC clock frequency from any CPU frequency above 100kHz. The prescaling is selected by the ADC Prescaler Select bits in the ADC Control and Status Register A (ADCSRA.ADPS). The prescaler starts counting from the moment the ADC is switched on by writing the ADC Enable bit ADCSRA.ADEN to '1'. The prescaler keeps running for as long as ADEN=1, and is continuously reset when ADEN=0.

When initiating a single ended conversion by writing a '1' to the ADC Start Conversion bit (ADCSRA.ADSC), the conversion starts at the following rising edge of the ADC clock cycle.

A normal conversion takes 13 ADC clock cycles. The first conversion after the ADC is switched on (i.e., ADCSRA.ADEN is written to '1') takes 25 ADC clock cycles in order to initialize the analog circuitry.

When the bandgap reference voltage is used as input to the ADC, it will take a certain time for the voltage to stabilize. If not stabilized, the first value read after the first conversion may be wrong.

The actual sample-and-hold takes place 1.5 ADC clock cycles after the start of a normal conversion and 13.5 ADC clock cycles after the start of an first conversion. When a conversion is complete, the result is written to the ADC Data Registers (ADCL and ADCH), and the ADC Interrupt Flag (ADCSRA.ADIF) is set. In Single Conversion mode, ADCSRA.ADSC is cleared simultaneously. The software may then set ADCSRA.ADSC again, and a new conversion will be initiated on the first rising ADC clock edge.

When Auto Triggering is used, the prescaler is reset when the trigger event occurs. This assures a fixed delay from the trigger event to the start of conversion. In this mode, the sample-and-hold takes place two

ADC clock cycles after the rising edge on the trigger source signal. Three additional CPU clock cycles are used for synchronization logic.

In Free Running mode, a new conversion will be started immediately after the conversion completes, while ADCRSA.ADSC remains high. See also the ADC Conversion Time table below.

Figure 28-4. ADC Timing Diagram, First Conversion (Single Conversion Mode)

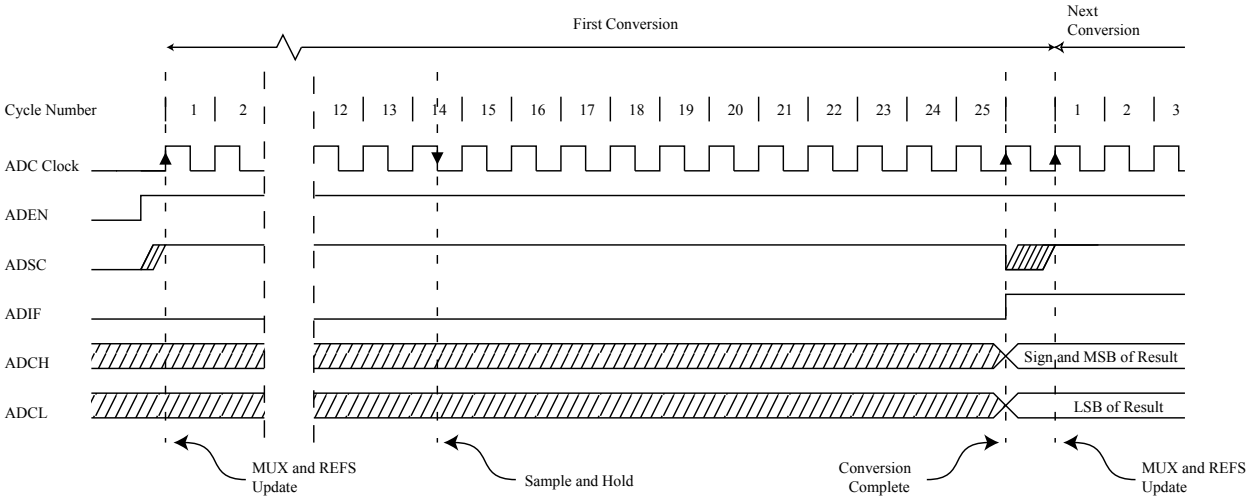


Figure 28-5. ADC Timing Diagram, Single Conversion

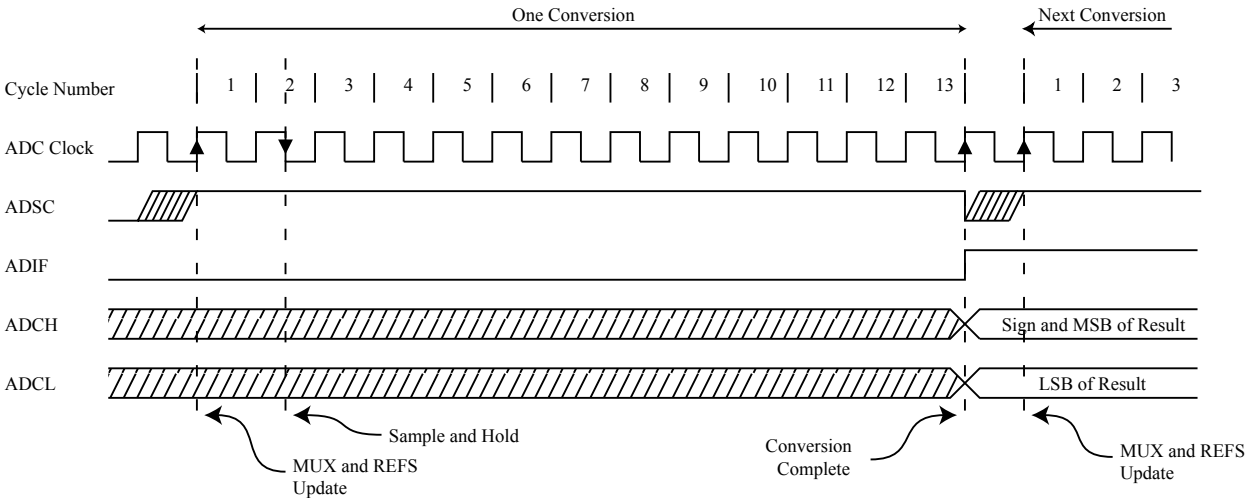


Figure 28-6. ADC Timing Diagram, Auto Triggered Conversion

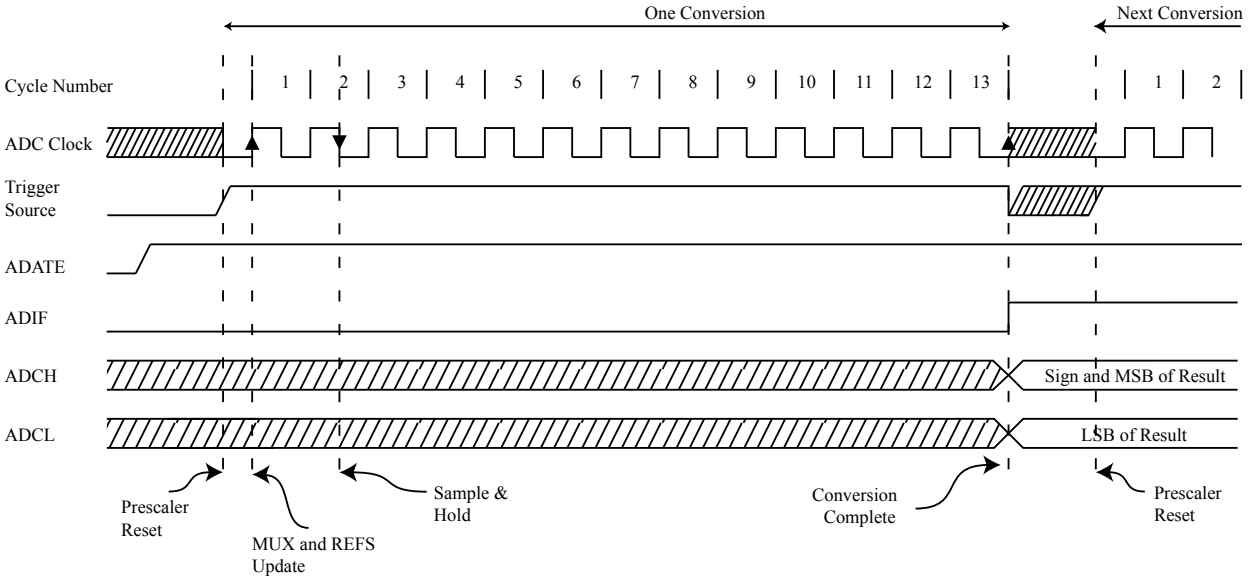


Figure 28-7. ADC Timing Diagram, Free Running Conversion

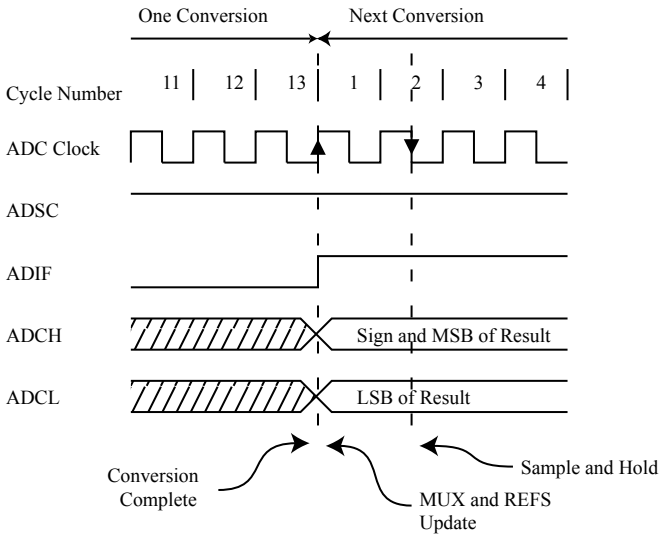


Table 28-1. ADC Conversion Time

Condition	Sample & Hold (Cycles from Start of Conversion)	Conversion Time (Cycles)
First conversion	13.5	25
Normal conversions, single ended	1.5	13
Auto Triggered conversions	2	13.5

28.5. Changing Channel or Reference Selection

The Analog Channel Selection bits (MUX) and the Reference Selection bits (REFS) bits in the ADC Multiplexer Selection Register (ADMUX.MUX[3:0] and ADMUX.REFS[1:0]) are single buffered through a temporary register to which the CPU has random access. This ensures that the channels and reference selection only takes place at a safe point during the conversion. The channel and reference selection is

continuously updated until a conversion is started. Once the conversion starts, the channel and reference selection is locked to ensure a sufficient sampling time for the ADC. Continuous updating resumes in the last ADC clock cycle before the conversion completes (indicated by ADCSRA.ADIF set). Note that the conversion starts on the following rising ADC clock edge after ADSC is written. The user is thus advised not to write new channel or reference selection values to ADMUX until one ADC clock cycle after the ADC Start Conversion bit (ADCRSA.ADSC) was written.

If Auto Triggering is used, the exact time of the triggering event can be indeterministic. Special care must be taken when updating the ADMUX Register, in order to control which conversion will be affected by the new settings.

If both the ADC Auto Trigger Enable and ADC Enable bits (ADCRSA.ADATE, ADCRSA.ADEN) are written to '1', an interrupt event can occur at any time. If the ADMUX Register is changed in this period, the user cannot tell if the next conversion is based on the old or the new settings. ADMUX can be safely updated in the following ways:

1. When ADATE or ADEN is cleared.
    1.1. During conversion, minimum one ADC clock cycle after the trigger event.
    1.2. After a conversion, before the Interrupt Flag used as trigger source is cleared.

When updating ADMUX in one of these conditions, the new settings will affect the next ADC conversion.

28.5.1. ADC Input Channels

When changing channel selections, the user should observe the following guidelines to ensure that the correct channel is selected:

- In Single Conversion mode, always select the channel before starting the conversion. The channel selection may be changed one ADC clock cycle after writing one to ADSC. However, the simplest method is to wait for the conversion to complete before changing the channel selection.
- In Free Running mode, always select the channel before starting the first conversion. The channel selection may be changed one ADC clock cycle after writing one to ADSC. However, the simplest method is to wait for the first conversion to complete, and then change the channel selection. Since the next conversion has already started automatically, the next result will reflect the previous channel selection. Subsequent conversions will reflect the new channel selection.

The user is advised not to write new channel or reference selection values during Free Running mode.

28.5.2. ADC Voltage Reference

The reference voltage for the ADC (V_{REF}) indicates the conversion range for the ADC. Single ended channels that exceed V_{REF} will result in codes close to 0x3FF. V_{REF} can be selected as either AV_{CC}, internal 1.1V reference, or external AREF pin.

AV_{CC} is connected to the ADC through a passive switch. The internal 1.1V reference is generated from the internal bandgap reference (V_{BG}) through an internal amplifier. In either case, the external AREF pin is directly connected to the ADC, and the reference voltage can be made more immune to noise by connecting a capacitor between the AREF pin and ground. V_{REF} can also be measured at the AREF pin with a high impedance voltmeter. Note that V_{REF} is a high impedance source, and only a capacitive load should be connected in a system.

If the user has a fixed voltage source connected to the AREF pin, the user may not use the other reference voltage options in the application, as they will be shorted to the external voltage. If no external voltage is applied to the AREF pin, the user may switch between AV_{CC} and 1.1V as reference selection. The first ADC conversion result after switching reference voltage source may be inaccurate, and the user is advised to discard this result.

28.6. ADC Noise Canceler

The ADC features a noise canceler that enables conversion during sleep mode to reduce noise induced from the CPU core and other I/O peripherals. The noise canceler can be used with ADC Noise Reduction and Idle mode. To make use of this feature, the following procedure should be used:

1. Make sure that the ADC is enabled and is not busy converting. Single Conversion mode must be selected and the ADC conversion complete interrupt must be enabled.
2. Enter ADC Noise Reduction mode (or Idle mode). The ADC will start a conversion once the CPU has been halted.
3. If no other interrupts occur before the ADC conversion completes, the ADC interrupt will wake up the CPU and execute the ADC Conversion Complete interrupt routine. If another interrupt wakes up the CPU before the ADC conversion is complete, that interrupt will be executed, and an ADC Conversion Complete interrupt request will be generated when the ADC conversion completes. The CPU will remain in active mode until a new sleep command is executed.

Note: The ADC will not be automatically turned off when entering other sleep modes than Idle mode and ADC Noise Reduction mode. The user is advised to write zero to ADCRSA.ADEN before entering such sleep modes to avoid excessive power consumption.

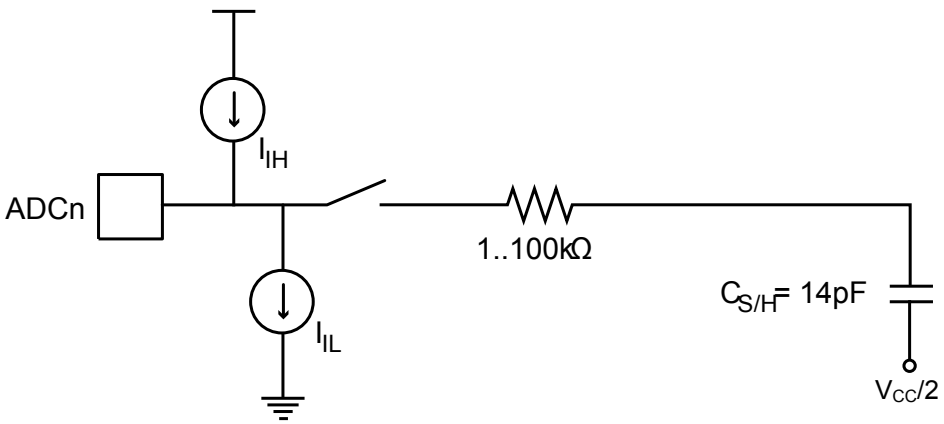
28.6.1. Analog Input Circuitry

The analog input circuitry for single ended channels is illustrated below. An analog source applied to ADCn is subjected to the pin capacitance and input leakage of that pin, regardless of whether that channel is selected as input for the ADC. When the channel is selected, the source must drive the S/H capacitor through the series resistance (combined resistance in the input path).

The ADC is optimized for analog signals with an output impedance of approximately 10 kΩ or less. If such a source is used, the sampling time will be negligible. If a source with higher impedance is used, the sampling time will depend on how long time the source needs to charge the S/H capacitor, with can vary widely. The user is recommended to only use low impedance sources with slowly varying signals, since this minimizes the required charge transfer to the S/H capacitor.

Signal components higher than the Nyquist frequency ($f_{ADC}/2$) should not be present for either kind of channels, to avoid distortion from unpredictable signal convolution. The user is advised to remove high frequency components with a low-pass filter before applying the signals as inputs to the ADC.

Figure 28-8. Analog Input Circuitry



28.6.2. Analog Noise Canceling Techniques

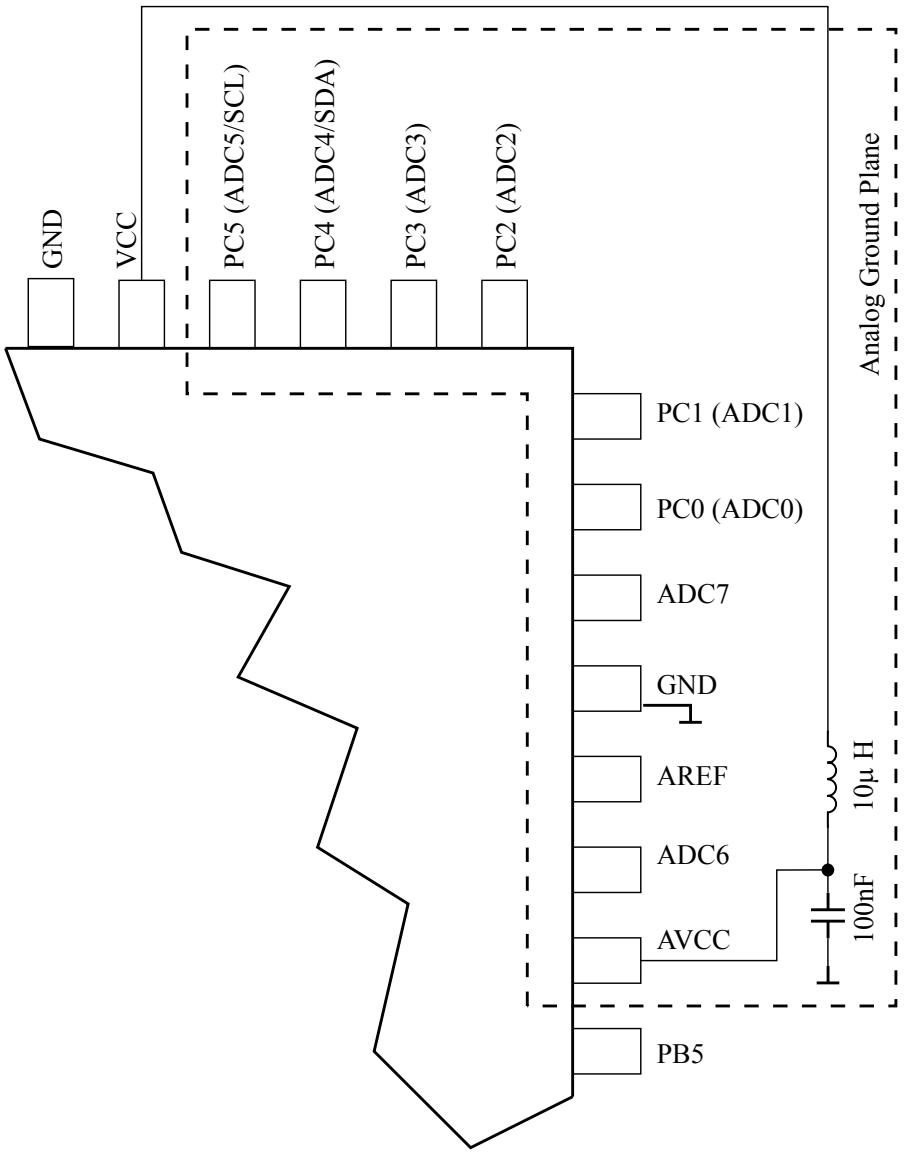
Digital circuitry inside and outside the device generates EMI which might affect the accuracy of analog measurements. If conversion accuracy is critical, the noise level can be reduced by applying the following techniques:

1. Keep analog signal paths as short as possible. Make sure analog tracks run over the analog ground plane, and keep them well away from high-speed switching digital tracks.
   1.1. The AV_{CC} pin on the device should be connected to the digital V_{CC} supply voltage via an LC network as shown in the figure below.

   1.2. Use the ADC noise canceler function to reduce induced noise from the CPU.

   1.3. If any ADC [3:0] port pins are used as digital outputs, it is essential that these do not switch while a conversion is in progress. However, using the 2-wire Interface (ADC4 and ADC5) will only affect the conversion on ADC4 and ADC5 and not the other ADC channels.

Figure 28-9. ADC Power Connections



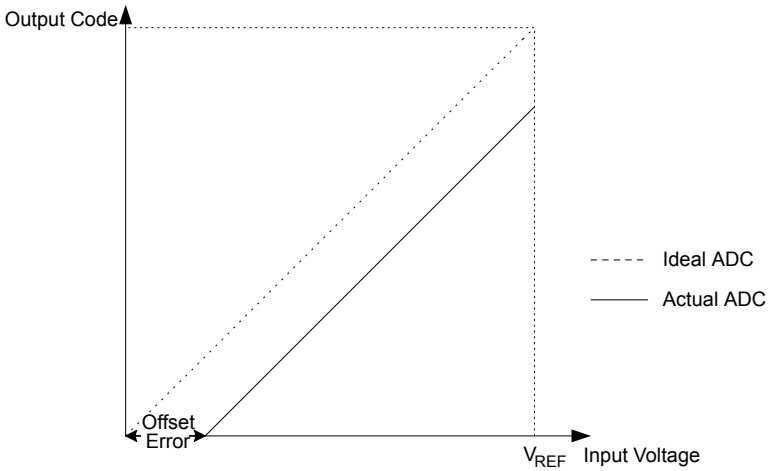
28.6.3. ADC Accuracy Definitions

An n-bit single-ended ADC converts a voltage linearly between GND and V_{REF} in 2^n steps (LSBs). The lowest code is read as 0, and the highest code is read as 2^n-1.

Several parameters describe the deviation from the ideal behavior:

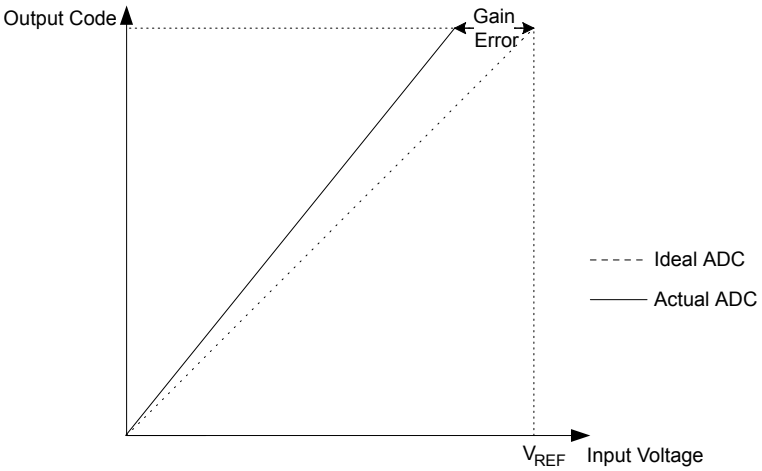
- Offset: The deviation of the first transition (0x000 to 0x001) compared to the ideal transition (at 0.5 LSB). Ideal value: 0 LSB.

Figure 28-10. Offset Error



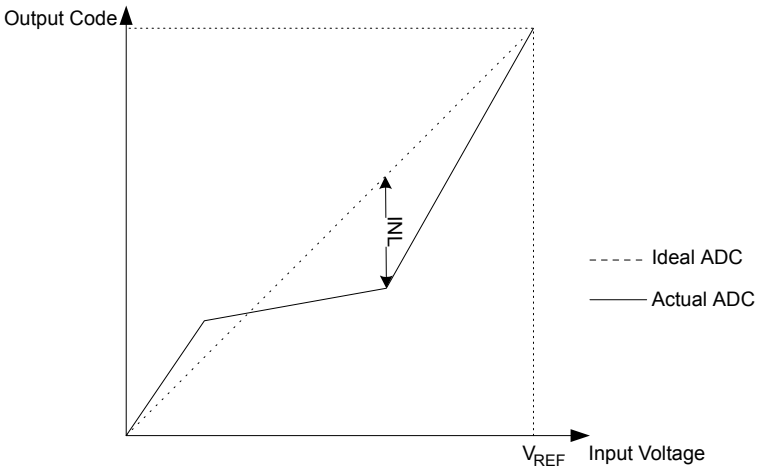
- Gain error: After adjusting for offset, the gain error is found as the deviation of the last transition (0x3FE to 0x3FF) compared to the ideal transition (at 1.5 LSB below maximum). Ideal value: 0 LSB.

Figure 28-11. Gain Error



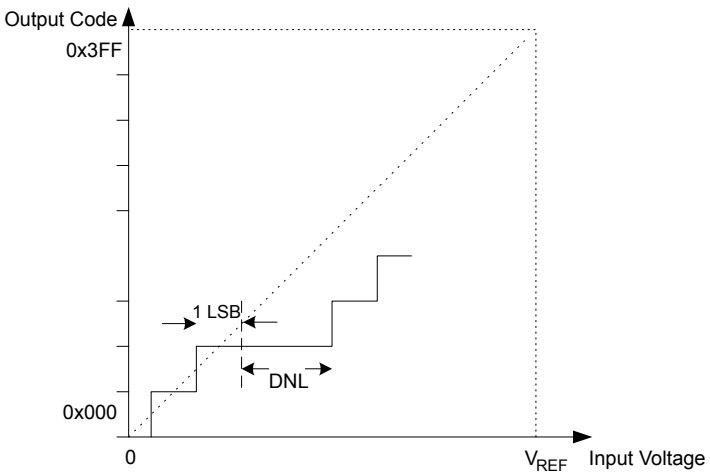
- Integral Non-linearity (INL): After adjusting for offset and gain error, the INL is the maximum deviation of an actual transition compared to an ideal transition for any code. Ideal value: 0 LSB.

Figure 28-12. Integral Non-linearity (INL)



- Differential Non-linearity (DNL): The maximum deviation of the actual code width (the interval between two adjacent transitions) from the ideal code width (1 LSB). Ideal value: 0 LSB.

Figure 28-13. Differential Non-linearity (DNL)



- Quantization Error: Due to the quantization of the input voltage into a finite number of codes, a range of input voltages (1 LSB wide) will code to the same value. Always ±0.5 LSB.
- Absolute accuracy: The maximum deviation of an actual (unadjusted) transition compared to an ideal transition for any code. This is the compound effect of offset, gain error, differential error, non-linearity, and quantization error. Ideal value: ±0.5 LSB.

28.7. ADC Conversion Result

After the conversion is complete (ADCSRA.ADIF is set), the conversion result can be found in the ADC Result Registers (ADCL, ADCH).

For single ended conversion, the result is

$$\mathrm{ADC} = \frac{V_{\mathrm{IN}} \cdot 1024}{V_{\mathrm{REF}}}$$

where V_{IN} is the voltage on the selected input pin, and V_{REF} the selected voltage reference (see also descriptions of ADMUX.REFSn and ADMUX.MUX). 0x000 represents analog ground, and 0x3FF represents the selected reference voltage minus one LSB.

Example:

ADMUX = 0xED (ADC3 - ADC2, 10× gain, 2.56V reference, left adjusted result)

Voltage on ADC3 is 300 mV, voltage on ADC2 is 500 mV.

ADCR = 512 × 10 × (300 - 500) / 2560 = -400 = 0x270

ADCL will thus read 0x00, and ADCH will read 0x9C.

Writing zero to ADLAR right adjusts the result: ADCL = 0x70, ADCH = 0x02.

28.8. Temperature Measurement

The temperature measurement is based on an on-chip temperature sensor that is coupled to a single ended Temperature sensor channel. Selecting the Temperature sensor channel by writing ADMUX.MUX[3:0] to '1000' enables the temperature sensor. The internal 1.1V voltage reference must also be selected for the ADC voltage reference source in the temperature sensor measurement. When the temperature sensor is enabled, the ADC converter can be used in single conversion mode to measure the voltage over the temperature sensor.

The measured voltage has a linear relationship to the temperature as described in the following table. The voltage sensitivity is approximately 1mV/°C, the accuracy of the temperature measurement is ±10°C.

Table 28-2. Temperature vs. Sensor Output Voltage (Typical Case)

Temperature	-45°C	+25°C	+85°C
Voltage	242mV	314mV	380mV

The values described in the table above are typical values. However, due to process variations the temperature sensor output voltage varies from one chip to another. To be capable of achieving more accurate results the temperature measurement can be calibrated in the application software. The software calibration requires that a calibration value is measured and stored in a register or EEPROM for each chip, as a part of the production test. The software calibration can be done utilizing the formula:

$T = \{ [(ADCH << 8) | ADCL] - T_{OS} \} / k$

where ADCH and ADCL are the ADC data registers, k is a fixed coefficient and T_{OS} is the temperature sensor offset value determined and stored into EEPROM as a part of the production test.

28.9. Register Description

28.9.1. ADC Multiplexer Selection Register

Name: ADMUX
Offset: 0x7C
Reset: 0x00
Property: -

Bit	7	6	5	4	3	2	1	0
	REFS1	REFS0	ADLAR		MUX3	MUX2	MUX1	MUX0
Access	R/W	R/W	R/W		R/W	R/W	R/W	R/W
Reset	0	0	0		0	0	0	0

Bits 7:6 – REFSn: Reference Selection [n = 1:0]
These bits select the voltage reference for the ADC. If these bits are changed during a conversion, the change will not go in effect until this conversion is complete (ADIF in ADCSRA is set). The internal voltage reference options may not be used if an external reference voltage is being applied to the AREF pin.

Table 28-3. ADC Voltage Reference Selection

REFS[1:0]	Voltage Reference Selection
00	AREF, Internal V_{ref} turned off
01	AV_{CC} with external capacitor at AREF pin
10	Reserved
11	Internal 1.1V Voltage Reference with external capacitor at AREF pin

Bit 5 – ADLAR: ADC Left Adjust Result
The ADLAR bit affects the presentation of the ADC conversion result in the ADC Data Register. Write one to ADLAR to left adjust the result. Otherwise, the result is right adjusted. Changing the ADLAR bit will affect the ADC Data Register immediately, regardless of any ongoing conversions. For a complete description of this bit, see ADCL and ADCH.

Bits 3:0 – MUXn: Analog Channel Selection [n = 3:0]
The value of these bits selects which analog inputs are connected to the ADC. If these bits are changed during a conversion, the change will not go in effect until this conversion is complete (ADIF in ADCSRA is set).

Table 28-4. Input Channel Selection

MUX[3:0]	Single Ended Input
0000	ADC0
0001	ADC1
0010	ADC2
0011	ADC3
0100	ADC4
0101	ADC5

MUX[3:0]	Single Ended Input
0110	ADC6
0111	ADC7
1000	Temperature sensor
1001	Reserved
1010	Reserved
1011	Reserved
1100	Reserved
1101	Reserved
1110	1.1V (V_{BG})
1111	0V (GND)

28.9.2. ADC Control and Status Register A

Name: ADCSRA
Offset: 0x7A
Reset: 0x00
Property: -

Bit	7	6	5	4	3	2	1	0
	ADEN	ADSC	ADATE	ADIF	ADIE	ADPS2	ADPS1	ADPS0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit 7 – ADEN: ADC Enable
Writing this bit to one enables the ADC. By writing it to zero, the ADC is turned off. Turning the ADC off while a conversion is in progress, will terminate this conversion.

Bit 6 – ADSC: ADC Start Conversion
In Single Conversion mode, write this bit to one to start each conversion. In Free Running mode, write this bit to one to start the first conversion. The first conversion after ADSC has been written after the ADC has been enabled, or if ADSC is written at the same time as the ADC is enabled, will take 25 ADC clock cycles instead of the normal 13. This first conversion performs initialization of the ADC.

ADSC will read as one as long as a conversion is in progress. When the conversion is complete, it returns to zero. Writing zero to this bit has no effect.

Bit 5 – ADATE: ADC Auto Trigger Enable
When this bit is written to one, Auto Triggering of the ADC is enabled. The ADC will start a conversion on a positive edge of the selected trigger signal. The trigger source is selected by setting the ADC Trigger Select bits, ADTS in ADCSRB.

Bit 4 – ADIF: ADC Interrupt Flag
This bit is set when an ADC conversion completes and the Data Registers are updated. The ADC Conversion Complete Interrupt is executed if the ADIE bit and the I-bit in SREG are set. ADIF is cleared by hardware when executing the corresponding interrupt handling vector. Alternatively, ADIF is cleared by writing a logical one to the flag. Beware that if doing a Read-Modify-Write on ADCSRA, a pending interrupt can be disabled. This also applies if the SBI and CBI instructions are used.

Bit 3 – ADIE: ADC Interrupt Enable
When this bit is written to one and the I-bit in SREG is set, the ADC Conversion Complete Interrupt is activated.

Bits 2:0 – ADPSn: ADC Prescaler Select [n = 2:0]
These bits determine the division factor between the system clock frequency and the input clock to the ADC.

Table 28-5. Input Channel Selection

ADPS[2:0]	Division Factor
000	2
001	2

ADPS[2:0]	Division Factor
010	4
011	8
100	16
101	32
110	64
111	128

28.9.3. ADC Data Register Low (ADLAR=0)

When an ADC conversion is complete, the result is found in these two registers.

When ADCL is read, the ADC Data Register is not updated until ADCH is read. Consequently, if the result is left adjusted and no more than 8-bit precision is required, it is sufficient to read ADCH. Otherwise, ADCL must be read first, then ADCH.

The ADLAR bit and the MUXn bits in ADMUX affect the way the result is read from the registers. If ADLAR is set, the result is left adjusted. If ADLAR is cleared (default), the result is right adjusted.

Name: ADCL
Offset: 0x78
Reset: 0x00
Property: ADLAR = 0

Bit	7	6	5	4	3	2	1	0
	ADC7	ADC6	ADC5	ADC4	ADC3	ADC2	ADC1	ADC0
Access	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bits 7:0 – ADCn: ADC Conversion Result [n = 7:0]
These bits represent the result from the conversion. Refer to ADC Conversion Result for details.

28.9.4. ADC Data Register High (ADLAR=0)

Name: ADCH
Offset: 0x79
Reset: 0x00
Property: ADLAR = 0

Bit	7	6	5	4	3	2	1	0
							ADC9	ADC8
Access							R	R
Reset							0	0

Bit 1 – ADC9: ADC Conversion Result
Refer to ADCL.

Bit 0 – ADC8: ADC Conversion Result

28.9.5. ADC Data Register Low (ADLAR=1)

Name: ADCL
Offset: 0x78
Reset: 0x00
Property: ADLAR = 1

Bit	7	6	5	4	3	2	1	0
	ADC1	ADC0						
Access	R	R						
Reset	0	0						

Bit 7 – ADC1: ADC Conversion Result
Refer to ADCL.

Bit 6 – ADC0: ADC Conversion Result

28.9.6. ADC Data Register High (ADLAR=1)

Name: ADCH
Offset: 0x79
Reset: 0x00
Property: ADLAR = 1

Bit	7	6	5	4	3	2	1	0
	ADC9	ADC8	ADC7	ADC6	ADC5	ADC4	ADC3	ADC2
Access	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bit 7 – ADC9: ADC Conversion Result 9
Refer to ADCL.

Bit 6 – ADC8: ADC Conversion Result 8
Refer to ADCL.

Bit 5 – ADC7: ADC Conversion Result 7
Refer to ADCL.

Bit 4 – ADC6: ADC Conversion Result 6
Refer to ADCL.

Bit 3 – ADC5: ADC Conversion Result 5
Refer to ADCL.

Bit 2 – ADC4: ADC Conversion Result 4
Refer to ADCL.

Bit 1 – ADC3: ADC Conversion Result 3
Refer to ADCL.

Bit 0 – ADC2: ADC Conversion Result 2
Refer to ADCL.

28.9.7. ADC Control and Status Register B

Name: ADCSRB
Offset: 0x7B
Reset: 0x00
Property: -

Bit	7	6	5	4	3	2	1	0
		ACME				ADTS2	ADTS1	ADTS0
Access		R/W				R/W	R/W	R/W
Reset		0				0	0	0

Bit 6 – ACME: Analog Comparator Multiplexer Enable
When this bit is written logic one and the ADC is switched off (ADEN in ADCSRA is zero), the ADC multiplexer selects the negative input to the Analog Comparator. When this bit is written logic zero, AIN1 is applied to the negative input of the Analog Comparator. For a detailed description of this bit, see *Analog Comparator Multiplexed Input*..

Bits 2:0 – ADTSn: ADC Auto Trigger Source [n = 2:0]
If ADATE in ADCSRA is written to one, the value of these bits selects which source will trigger an ADC conversion. If ADATE is cleared, the ADTS[2:0] settings will have no effect. A conversion will be triggered by the rising edge of the selected Interrupt Flag. Note that switching from a trigger source that is cleared to a trigger source that is set, will generate a positive edge on the trigger signal. If ADEN in ADCSRA is set, this will start a conversion. Switching to Free Running mode (ADTS[2:0]=0) will not cause a trigger event, even if the ADC Interrupt Flag is set.

Table 28-6. ADC Auto Trigger Source Selection

ADTS[2:0]	Trigger Source
000	Free Running mode
001	Analog Comparator
010	External Interrupt Request 0
011	Timer/Counter0 Compare Match A
100	Timer/Counter0 Overflow
101	Timer/Counter1 Compare Match B
110	Timer/Counter1 Overflow
111	Timer/Counter1 Capture Event

28.9.8. Digital Input Disable Register 0

When the respective bits are written to logic one, the digital input buffer on the corresponding ADC pin is disabled. The corresponding PIN Register bit will always read as zero when this bit is set. When an analog signal is applied to the ADC7...0 pin and the digital input from this pin is not needed, this bit should be written logic one to reduce power consumption in the digital input buffer.

Name: DIDR0
Offset: 0x7E
Reset: 0x00
Property: -

Bit	7	6	5	4	3	2	1	0
	ADC7D	ADC6D	ADC5D	ADC4D	ADC3D	ADC2D	ADC1D	ADC0D
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 0, 1, 2, 3, 4, 5, 6, 7 – ADC0D, ADC1D, ADC2D, ADC3D, ADC4D, ADC5D, ADC6D, ADC7D: ADC Digital Input Disable